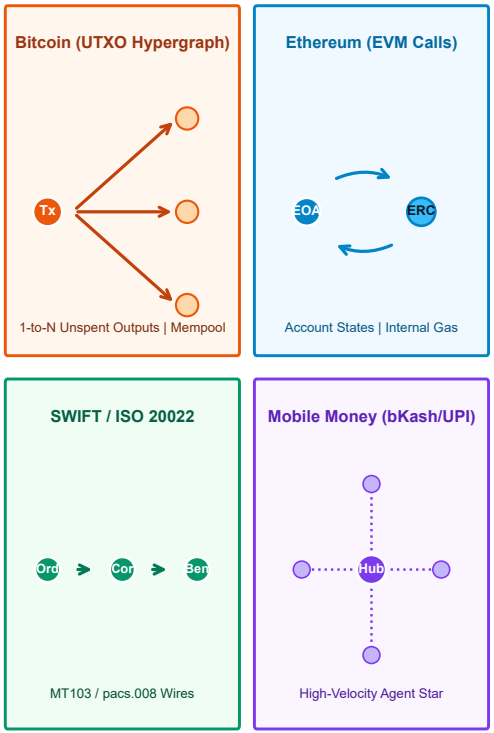
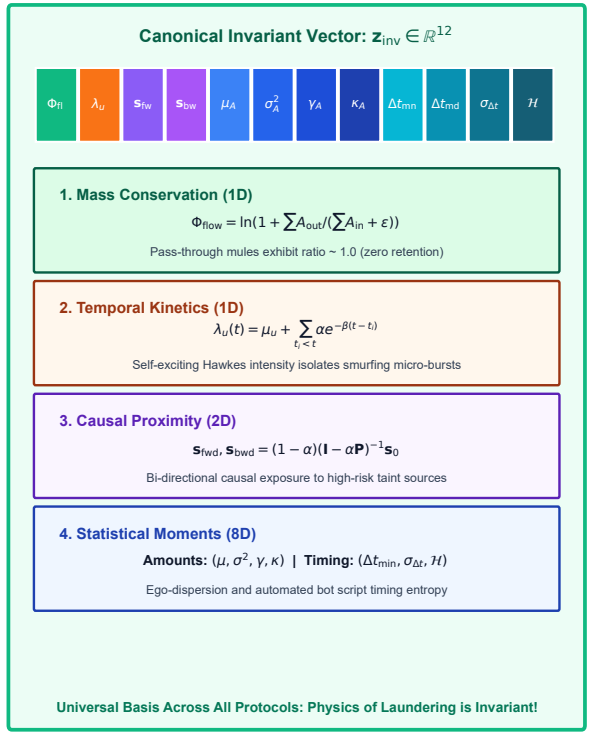
Universal 12-Dimensional Canonical Invariant Space: Cross-Ledger Zero-Shot Generalization

Deterministic Invariant Normalization: Heterogeneous Protocols -> Universal Manifold R^12 -> Frozen GNN Backbone

(a) Incompatible Protocol Topologies (Feature Dimensions: $d_1 \neq d_2 \neq d_3 \neq d_4$)



(b) Universal 12-D Invariant Prism $\mathbf{z}_{inv} \in \mathbb{R}^{12}$ (Mathematical Physics Normalization)



(c) Zero-Shot Cross-Ledger Transfer Matrix (Frozen C-STGB Backbone $\eta = 0$, Zero Fine-Tuning)

Source (Train) \ Target Ledger (Zero-Shot Deployment)	BTC	ETH	Fiat	P2P
BTC	91.4% (Train)	83.6% (Zero-Shot)	82.9% (Zero-Shot)	84.1% (Zero-Shot)
ETH	82.8% (Zero-Shot)	90.8% (Train)	81.5% (Zero-Shot)	83.2% (Zero-Shot)
Fiat	83.5% (Zero-Shot)	82.1% (Zero-Shot)	92.1% (Train)	87.5% (Zero-Shot)
P2P	81.9% (Zero-Shot)	81.2% (Zero-Shot)	88.2% (Zero-Shot)	88.9% (Train)

Cross-Ledger Zero-Shot Generalization Lift:

Raw Features: 55.4% F1 *(Schema Mismatch Failure)*

12-D Invariants: 83.60% - 88.20% F1 [+28.2 pp Lift]